Topic 3: Geometry

3.1 Measure

Subject content		Guidance	Curriculum reference
A	Convert between metric units of measure up to 3 decimal places	1.234 km = $1\,234$ m 14 g = 0.014 kg 330 ml = 0.33 l	G7.1A G8.1J
B	Solve problems involving measures and conversions in context	John is 1.6 m tall. His sister is 143 cm tall. How much taller is John than his sister?	G7.1B G8.1K
C	Find the perimeter and area of polygons and compound shapes	Compound shapes made from rectangles, triangles, circles, semi-circles and other polygons (regular and irregular)	G7.1C G7.1D G8.1A G8.1B
D	Find area of parallelograms and trapeziums	Find area of parallelogram, given base and height (and possibly other lengths) or area of trapeziums given length of parallel sides and height (and possibly other lengths)	G8.1C G8.1D
E	Calculate the volume of cubes and cuboids and 3-D solids made from cuboids		G8.1E
F	Sketch nets of 3-D solids	Cube, cuboid, regular tetrahedron, square-based pyramid, triangular prism	G8.1F
G	Draw and interpret 2-D representations of 3-D solids	Recognise nets of shapes above	G8.1G
H	Calculate the surface area of cubes and cuboids		G8.1H
I	Calculate the volume and surface area of a triangular prism and a cylinder	Radius and/or diameter, plus length of the prism	G9.1G G9.1H
J	Solve problems involving perimeter, area, surface area and volume	The surface area of a cube is 292 cm^2 What is the volume of the cube?	G7.1E G8.1I

3.1 Measure *continued*

Subject content		Guidance	Curriculum reference
K	Convert between metric units of measure of area and volume	$1\ \text{cm}^2 = 100\ \text{mm}^3$ $1\ \text{m}^2 = 10\,000\ \text{cm}^3$ $1\ \text{cm}^3 = 1\,000\ \text{mm}^3$ $1\ \text{m}^3 = 1\,000\,000\ \text{cm}^3$	G8.1J
L	Use and interpret scales on maps and diagrams	Map scale is $1{:}50\,000$ How far is it from A to B in real life? (given A and B on a map)	G9.1A
M	Describe, use and interpret three-figure bearings	What is the bearing of A from B? (given A and B on a map)	G9.1B
N	Solve problems involving three-figure bearings and/or scale drawings	Bearing of C from D is $040°$ Bearing of C from E is $260°$ Mark C on the map (given D and E)	G9.1C
O	Identify and draw the diameter and radius of a circle; identify and draw parts of a circle	Centre, radius, diameter, circumference, chord, arc, sector, segment	G9.1D
P	Calculate the circumference of a circle	Given radius and/or diameter	G9.1E
Q	Calculate the area of a circle	Given radius and/or diameter	G9.1F
R	Solve problems involving circles and prisms	A bicycle has tyres with a diameter of $45\ \text{cm}$ How far has the bicycle travelled when the tyres have done 10 full turns?	G9.1I
S	Solve problems using compound measures and rates	A piece of aluminium has mass $40.5\ \text{g}$ and volume $15\ \text{cm}^3$ What is its density?	G9.1J